

The BE WELL study: Black Raspberry Beverage Working to Prevent Lung Cancer

Pilot study to define compliance and impact of a novel black raspberry food product on the microbiome and immune functions in a high risk cohort for lung cancer.

Principal Investigators:

Daniel Spakowicz, PhD, MS

Research Assistant Professor
Division of Medical Oncology
Department of Internal Medicine &
Department of Biomedical Informatics
The Ohio State University Comprehensive Cancer Center
482 Biomedical Research Tower
460 West 10th Ave.
Columbus, OH 43210
Phone: (614) 293-1797
Email: Daniel.Spakowicz@osumc.edu

David Carbone, MD, PhD

Professor
Division of Oncology
Department of Internal Medicine
The Ohio State University College of Medicine
488 Biomedical Research Tower
460 West 10th Ave.
Columbus, OH 43210
Phone: (614) 685-4479, Fax: (614) 293-4372
Email: david.carbone@osumc.edu

Steven K. Clinton, MD, PhD

Professor
Division of Medical Oncology
Department of Internal Medicine
The Ohio State University College of Medicine
A456 Starling Loving Hall
320 West 10th Ave.
Columbus, OH 43210
Phone: 614-293-2886, Fax: 614-293-7525
Email: steven.clinton@osumc.edu

Co-Investigators:

Marisa Bittoni

Research Scientist
The Ohio State University Comprehensive Cancer Center
473 Biomedical Research Tower
460 West 12th Ave.
Columbus, OH 43210
Email: Marisa.Bittoni@osumc.edu

Xiakui Mo

Research Scientist
Center for Biostatistics
The Ohio State University
320D 1800 Cannon Drive
Columbus, OH, 43210
Phone: (614) 293-3177
Email: Xiaokui.Mo@osumc.edu

Kenneth M. Riedl, PhD

Research Scientist
Department of Food Science and Technology
College of Food, Agricultural, and Environmental Sciences
The Ohio State University
Parker Food Science and Technology Building
2015 Fyffe Court
Columbus, OH 43210
Phone: 614-247-7686, Fax: 614-292-
Email: kenneth.riedl@osumc.edu

Yael Vodovotz, PhD

Associate Professor
Department of Food Science and Technology,
College of Food, Agricultural, and Environmental Sciences
The Ohio State University
Parker Food Science and Technology Building
2015 Fyffe Court
Columbus, OH 43210
Phone: 614-247-7696, Fax: 614-292-
Email: vodovotz.1@osu.edu

Gretchen Whitby

Nurse Practitioner
OSU Physicians, Inc
410 W 10th Avenue
Columbus, OH 43210
Phone: (614) 293-2878
Email: Gretchen.Whitby@osumc.edu

Michael Wert, MD

Assistant Professor-Clinical
Ste 2200, 2050 Kenny Rd
Columbus, OH-43221
Phone:(614) 293-4925
Fax: (614) 293-5503
Michael.Wert@osumc.edu

Filiz Oezkan, MD

Postdoctoral Researcher/ Visiting Scholar
Division of Medical Oncology
The Ohio State University College of Medicine
460 W. 12th Ave.
Columbus, OH 43210
Email: filiz.oezkan@osumc.edu

External Collaborators:

Rebecca Cloyes

Nurse Practitioner

OSU Physicians, Inc

410 W 10th Avenue

Columbus, OH 43210

Phone: (614) 293-6939

Email: Rebecca.Cloyes@osumc.edu

Rationale for the current amendment (v6, 2022-02-16):

1. Major
 - a. Blood-based analyses were updated to include an estimation of immune senescence. No additional samples are collected from study participants.
2. Minor
 - a. formatting changes to wrap text around figure 2
 - b. formatting changes to ensure consistency with how heading level 4 is displayed

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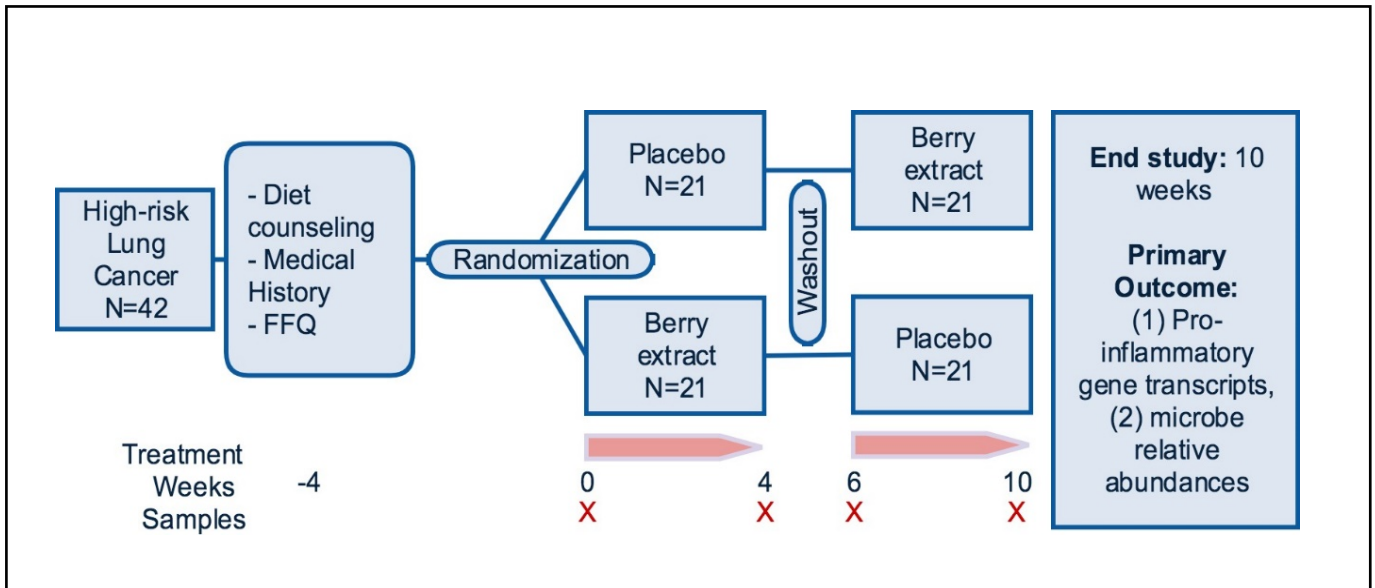


Figure 1. Study Schema.

Patients at high-risk for lung cancer (n=96) will be recruited and undergo baseline testing (diet counseling, medical history, Food Frequency Questionnaire (FFQ)) within 4 weeks of starting the intervention. Estimating a 56% drop-out rate before the first study visit, we expect 42 participants will proceed to randomization. Participants will be randomized to 1 of 2 groups: the Placebo First Group (N=21) and a Berry First Group (N=21). Participants will consume either a Placebo beverage or a Berry nectar beverage made from black raspberries (BRB) 2 x daily for a total of 4 weeks. All participants will undergo a Washout period between Weeks 4 and 6 and will then crossover and receive either the Berry Nectar or the Placebo, depending on their initial randomization, for weeks 6-10. At Week 10, the primary outcomes of the study are the (1) pro-inflammatory gene transcripts and the (2) microbe relative abundances.

1. Introduction

1.1 Lung cancer prevention strategies are needed

In 2018, approximately 234,030 new cases of lung cancer will occur in the United States¹. Despite major advances in recent years for screening and treatment, lung cancer remains the leading cause of cancer-related mortality, with approximately 154,050 deaths occurring in 2018 as a result of lung cancer. Five year survival rates range from 49% in local disease to just 2% in distant-stage disease¹. Given the poor survival rates in advanced stage disease, there is a critical need for improved prevention, early detection efforts, and studies focusing on modifiable lifestyle risk factors.

1.2 Rational manipulation of the microbiome as a strategy for preventing lung cancer

Targeting the microbiome may be a promising approach to reduce cancer risk and improve treatment outcomes in lung cancer. Although the lung has been thought to be sterile, recent studies have highlighted the role of the microbiome in the lung diseases and lung cancer^{2,3}. For instance, a significant relationship has been observed between *mycobacterium tuberculosis* (TB) and lung cancer in several studies^{4,5}. This link may be related to chronic inflammation-related carcinogenesis, given that microbes can be pro- or anti-inflammatory, resulting in changes in cancer risk^{6,7}. However, genotoxic effects, altered metabolism, and immune response are also being investigated as mediators of this link. Global changes of the microbiome, such as reduced alpha diversity (i.e. richness and evenness of the taxa within a sample) has also been related to these effects^{8,9}.

The microbiome has also been related to change in immune cell composition and circulation in the context of lung cancer¹⁰⁻¹². Important immune cells include regulatory T (Tregs) cells and type II Natural Killer T (NKT) cells, which suppress an anti-tumor immune response, and type I NKT cells promote an anti-tumor immune response. In preclinical experiments using mouse spleen cells, Berzofsky et al. showed that type II NKT cells produce IL-13 which causes myeloid suppressor cells (MDSCs) to produce TGF- β . This leads to the upregulation of regulatory T cells, which then suppress effector T cells. On the other hand, type I NKT cells upregulate effector T cells¹³. Collectively this suggests that the relative abundances of type I and type II NKT cells are a critical determinant of systemic inflammation and cancer surveillance. The abundance of NKT cells in human peripheral blood accounts for about 1-2% of lymphocytes, the majority are type II NKT cells (PMID 26895635, PMID 29520281). Since the abundance of type I NKT cells in human peripheral blood is very low, it is necessary to expand them to create positive controls for flow cytometry-based experiments. Furthermore, we are planning to use an NKT cell suppression assay, developed from our well-established Treg suppression protocol and an existing well-established NKT cell expansion protocol, to prove the immunosuppressive role of type II NKT cells in metastatic NSCLC patients, as compared to patients enrolled in this study and healthy controls.

1.3 Dietary intervention as a tool to modify the microbiome

Previous studies have employed probiotic supplements or fecal microbiota transplant (FMT) to alter the microbiome in attempt to mitigate cancer risk; however, there are disadvantages to these methods. First, probiotics have recently been shown to reduce diversity of the gut microbiome, which is associated with microbial benefits¹⁴. Second, for FMT as a potential treatment, some patients have had severe immune reactions, as it is unclear how best to identify whether donor stool is “safe” for transplant¹⁵.

Diet can also modify the microbiome¹⁶, and may, therefore, represent an approach that alleviates some of the issues of probiotic and FMT interventions. In fact, diets rich in fruits and vegetables are strongly linked with a lower risk of malignancy and tobacco-related cancers, such as lung cancer. Balansky et al. showed that the administration of aqueous berry nectars inhibited the cigarette-smoking induced formation of lung tumors in mice, as well as pulmonary emphysema, liver degeneration, weight loss and cytogenetical damage¹⁷. The biomarker C-reactive protein (CRP) was significantly elevated in the mice, which provides justification for testing in humans, following CRP levels and the expression of other pro-inflammatory biomarkers. In addition to mitigating cancer risk, findings indicate that optimal nutritional status can have an important impact on

response to immunotherapy in cancer patients¹⁶. A diet-based intervention to impact the microbiome may contribute positively to both cancer prevention and treatment outcomes.

1.4 Black Raspberry (BRB) nectar and anti-cancer effects

Berries contain multiple bioactive components including anthocyanins, which have demonstrated antiproliferative activity of lung cancers both *in vivo* and *in vitro*^{18,19}. Intriguingly, very little of these compounds are absorbed by the gut until they are modified by bacterial enzymes such as α -rhamnosidases, suggesting a role for the microbiome in mediating the effects^{20,21}. Moreover, these compounds have been shown to enrich certain microbial taxa such as *Akkermansia muciniphila*. This microbe has demonstrated positive effects in a variety of phenotypes, including obesity and diabetes^{22,23} and, more recently, has shown a causal role in promoting response to immune checkpoint inhibitors in a melanoma mouse model¹⁰.

A food-based approach has the potential to influence multiple targets relating to carcinogenesis with reduced toxicity. Studies pioneered at The Ohio State University (OSU) using BRB have shown positive results of BRB to inhibit cancerous processes^{24–28}. Early studies examined absorption and metabolism, following consumption of freeze dried powder dispersed in water^{27,28}. Others have used a pharmacologic approach of incorporating BRB components into bioadhesive gels and examined phytochemical exposure, alteration of metabolic enzymes, gene methylation and expression^{24–26}. A Phase I study of BRB colon cancer documented significant changes in epigenetic modifications and gene expression²⁷. Furthermore, original team member and principal investigator (PI), Dr. Steven Clinton and colleagues, recently completed studies with BRB candies in cohorts of oral cancer patients to assess molecular mechanisms of action²⁹. Together, these studies uniformly suggest that BRBs are bioactive and safe at the proposed dose.

1.5 Rationale for the proposed trial

Our research team has documented experience in developing and characterizing novel phytochemical rich food products for cancer prevention or as an adjunct to cancer therapy and has contributed to the development of BRB nectar as a whole food delivery system for cancer studies^{19,29–32}. BRB is fully characterized with a consistent and stable array of bioactive phytochemicals. Consumption of BRB, therefore, allows multiple bioactives to act upon an assortment of pathways to have greater efficacy and safety compared to a single drug. Additionally, there is evidence for anticancer effects and alteration of the microbiome, which may mediate the beneficial effects of the bioactive compounds of BRB²⁰. We, therefore, propose a randomized controlled trial to longitudinally monitor the microbiomes of high-risk lung cancer patients, as they are given a dietary intervention or placebo and to assess whether changes in the microbiomes are associated with systemic effects in patients' blood.

We predict that the berry nectar increases the relative abundance of the microbe *Akkermansia muciniphila*, which has been shown to induce response to immunotherapy in a mouse model whose microbiome was colonized by that of a human who did not respond to treatment¹⁰. In an unrelated study, it was shown that a diet rich in polyphenols, such as black raspberries, enrich for *A. muciniphila*²¹. However, no study has simultaneously monitored the microbiome and markers of lung cancer prevention in the context of these diet interventions. The approved study OSU-14135 is currently pioneering a BRB dietary intervention at the OSU Comprehensive Cancer Center (OSUCCC) for oral cancer where it assesses the oral microbiome. We use the same materials as this study, and our outcomes will sufficiently similar to integrate the results of the other study. Here, we plan to add more data to show that this intervention is also feasible and efficacious in a population of individuals at high risk for lung cancer. The results of this study will also add valuable information to the ongoing OSU-14135 study.

2. Objectives

Our central hypothesis is that phytochemicals in black raspberries enhances diversity of the gut microbiome, alters specific populations, and leads to decreased systemic and upper airway inflammation. We propose primary objectives in the categories of feasibility and science.

2.1 Primary feasibility objective

<i>Primary feasibility objective</i>	To evaluate the feasibility of establishing a diet intervention trial with longitudinal microbiome collection in OSUCCC Lung Cancer Screening Clinic (OSUCCC-LCSC)
<i>Primary feasibility endpoint</i>	Feasibility will be defined as the proportion of screened patients who agree to participate, study compliance: following low polyphenol diet, consumption of 2 x 170 mL BRB nectar daily, return all stool and urine samples to clinic, attend four additional clinic visits with successful blood draws and nasal swabs

2.2 Primary scientific objective

<i>Primary therapeutic objective</i>	To determine the impact of the BRB nectar intervention on the microbiome and inflammatory biomarkers.
<i>Primary therapeutic endpoints</i>	1) A significant correlation between microbes and inflammatory genes 2) A significant difference in microbial metabolic pathway abundances between treatment and placebo

3. Study Design and Overview

This randomized, placebo-controlled diet intervention crossover design trial is planned to determine whether dietary intervention and biospecimen collection is acceptable in this patient population. We plan to accrue a total of 42 participants who will be instructed to follow a “controlled phenolic diet” (**Appendix 1: Low Phenolic Diet Compliance**). Each participant will be randomized to the Placebo First or BRB First group. All participants will receive either 10mg of placebo or 10mg of BRB nectar to be consumed twice daily for Weeks 0-4, depending on their group of randomization. Week 4-6 will comprise a Washout Period in which all participants will not consume either beverage to eliminate possible carry-over effects in our crossover design (the wash-out period was determined based on the half-life of BRB nectar³⁰). After the Washout Period, participants will crossover to receive the other treatment, either placebo or BRB nectar, for weeks 6-10.

- Placebo First Group: Participants will first consume 10mg of a placebo (the placebo looks and tastes similar to the BRB nectar, but lacks the phytochemicals of berry nectar) twice daily for Weeks 0-4; they will consume the BRB nectar twice daily for Weeks 6-10.
- BRB First Group: Participants will first consume the BRB nectar twice daily for Weeks 0-4; they will consume the placebo twice daily for Weeks 6-10.

Samples of nasal brushings, blood, urine, and stool will be collected at Weeks 0, 4, 6, and 10 (see Section 5.3).

3.1 Justification for the trial design

This trial takes advantage of the unique nature of the high risk lung screening clinic, in that the patient population has elevated systemic inflammation from years of smoking but their phenotypes are relatively stable³³. By contrast, a similar trial in a cancer population would contend with confounding effects of the cancer and associated therapies. This allows for a crossover type trial design, whereby each individual is able to serve as their own control. The variation is therefore greatly minimized, which is critical in microbiome studies that show strong inter-individual variability. We have demonstrated this in previous microbiome studies that involve a diet intervention; each individual more closely resembled themselves before the diet intervention than they

resembled others in the trial post-intervention²².

The crossover design has several advantages: (1) It controls for any covariate imbalances, which often occur in parallel designs even after randomization. (2) They are statistically efficient, because the within-subject coefficient of variation (CV) is often 4X lower than between subject³⁴. We have shown this to be especially critical in the context of microbiome studies, where the zero-inflated negative binomial distributions make power estimates sensitive to the CV, and where microbe relative abundances can make the between-subjects coefficient of variation quite high²². This is described in Figure 2, where the power to detect a change in a microbial community member (negative binomial distribution, $\alpha = 0.05$, correcting for 200 tests) drastically changes over a 4X range in CV. This design is enabled by the patient population: smokers or former smokers with chronic, stable, high levels of inflammation in OSUCCC-LCSC.

Implicit in this trial design is the personalized nature of using each individual as their own control. We support the need for personalized medicine, and trial designs that speak to this need whenever possible.

3.1.1 Justification for the sample size.

Given the CV expected for the crossover design (0.3), 26 participants are needed to achieve 80% power to detect a 1.8 fold change in inflammatory transcripts after correcting for multiple hypothesis tests (**Figure 2**). Assuming a conservative compliance of 0.6 (See 2.1 Primary Feasibility Outcome), 42 need to be enrolled in the study. In addition to low compliance, we also expect a high dropout rate after consent and before the collection of the first sample. This is estimated at 56% from the first 50 participants. Therefore we expect to require consent of 96 individuals, with 42 proceeding to study visits and 26 completing all four study visit collections.

3.1.2 Justification for the treatment duration

A significant concern is that the duration of treatment will be insufficient to observe effects in inflammatory markers, particularly given the duration of insult (30 pack-year smoking history) that led patients to the LCSC. Duration was estimated by two previous reports that monitored similar changes following BRB intervention in different populations. Significant changes in plasma levels of GM-CSF and IL-8 were reported in patients consuming berries for more than 10 days and changes correlated with apoptosis induction and reduced proliferation in colorectal tissues³⁵. These results support that BRB induce changes in circulatory markers rather rapidly (10 days), but that tissue specific alterations require a longer duration of BRB administration, calculated as 4 or more weeks, or higher cumulative dose³⁶. We, therefore, conservatively chose both the longer estimate (4 weeks) as the duration of treatment, as well as a higher cumulative dose (2x daily, as opposed to 1x daily).

3.1.3 Justification for the washout period duration

A significant concern in any crossover trial design is the carry-over effect, whereby the treatment effects from the first intervention affect the results of the second intervention. We chose the two-week washout to conservatively control for possible carry-over effects in our crossover design. Previous studies have demonstrated the loss of dietary polyphenols from the urine 24 hours after ending consumption, and a return to baseline serum biomarkers levels within 72 hours³⁰. Therefore, conservatively estimating the serum half-life of BRB nectar-derived changes to be 48 hours, we chose a 2 week washout to allow 7 half-lives duration, or < 1% signal remaining. To further reduce the risk of carryover effects, we will use a block randomization scheme to randomize covariates we have shown to affect inflammatory markers, smoking status and sex³⁷, across the “treatment first” and “treatment second” arms of the cross-over, described in more detail in Section 5.1: Screening and Instructional Session.

4. Participant Selection

The study will recruit 42 eligible patients from OSUCCC-LCSC, which provides screening for high-risk patients. (See **Appendix 4: Eligibility Screener**).

4.1 Eligibility criteria

4.1.1 Inclusion Criteria: All patients will be selected from high-risk patients at the OSUCCC-LCSC. The process of being accepted into the lung screening clinic includes meeting the following criteria established by the US Preventive Services Task Force ³⁸:

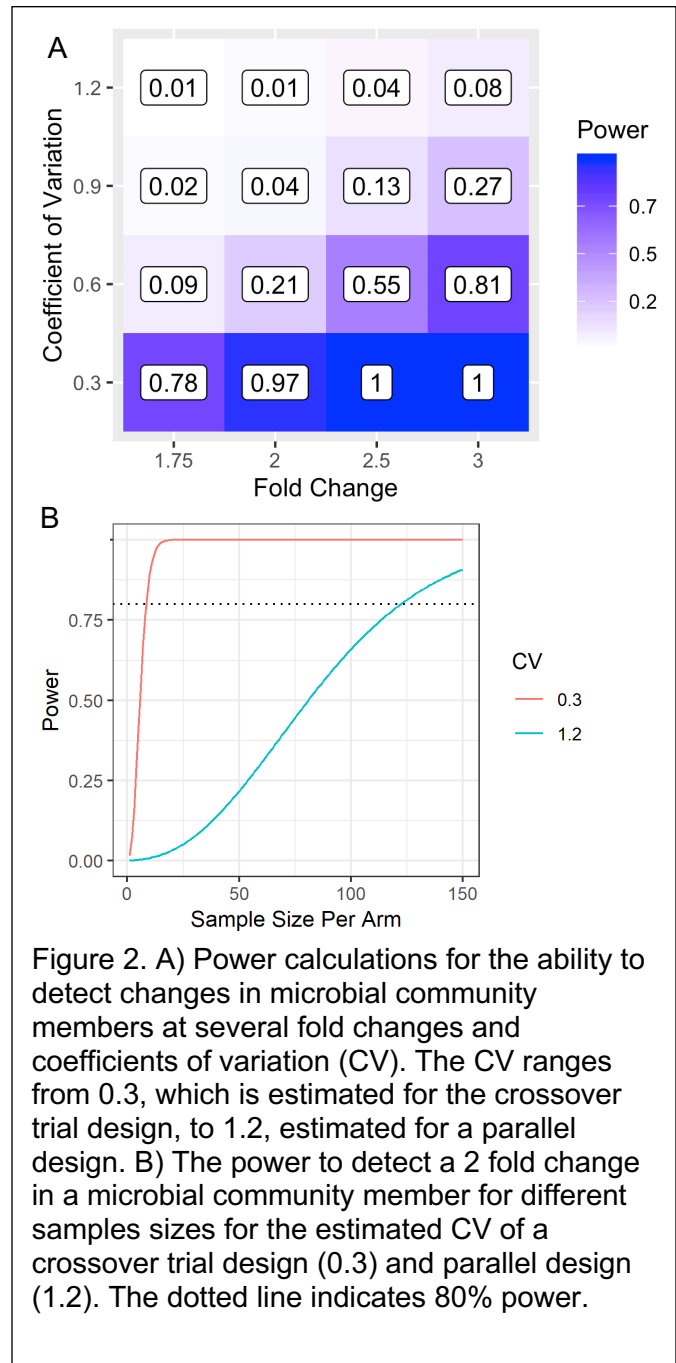
- Between 55 and 77 years of age.
- Have a 30 pack-year smoking history.
- Be either a current smoker or a former smoker that quit in the last 15 years. Any individual who is currently smoking and has smoked more than 100 cigarettes in their lifetime will be identified as a current smoker.

The additional criteria that need to be met as part of this study are that the participant must:

- Agree to consume a standardized vitamin / mineral supplement and avoid other nutritional, dietary or alternative medications / supplements / probiotics for the duration of the study.
- Agree to follow a berry-free / controlled phenolic diet (**Appendix 1: Low Phenolic Diet Compliance**) and to document consumption of polyphenolic foods each day of the study using a simple daily form.

4.1.2 Exclusion Criteria: Participants will be excluded from this study if:

- Person is <55 or >77 years of age
- Person has an allergy to any berries, pectin, or food colorants apples, as well as kiwi fruit, strawberries, soy sauce, pine nuts, almonds, cherries, peaches, blackberries, pears
- Person states that they are not a:
 - Current smoker: active cigarette smoker who has smoked more than 30 pack-years in the last 15 years OR
 - Former Smoker: not a current smoker active cigarette smoker who has smoked more than 30 pack years in the last 15 years
- Person is unwilling to follow a berry-free/controlled polyphenol diet while on study.
- Person currently has an uncontrolled metabolic disorder (uncontrolled diabetes, hyper- or hypothyroidism, etc.) or digestive illness that affects nutrient absorption (Crohn's disease, Celiac, renal/hepatic insufficiency, short bowel, etc.)
- Person has allergy or food intolerance to ingredients in study products (black raspberries or other berries)



- Person is on a regimen of any of the following medications:
 - Immunosuppressants, bisphosphonates, or steroids.
 - Anticoagulants (warfarin, apixaban, dabigatran, and rivaroxaban)
 - Probiotics
- Person is undergoing treatment for cancer in any form.
- Person is currently pregnant or nursing or plans to become pregnant during this study.
- Person plans to enter smoking cessation or change their smoking status during the course of the study.

4.2 Participant Recruitment

Recruitment will occur at the OSUCCC-LCSC Martha Morehouse Medical Plaza site. A study flyer will be placed in the clinic (**Appendix 3: Clinic Flyer**). The clinic is recognized by the American College of Radiology as a screening center of excellence and provides onsite shared decision-making, Low-Dose Computed Tomography (LDCT) and real time interpretation of test results. During clinic hours, a thoracic radiologist is available for CT interpretation in real time (within 30 minutes) and all results are provided in our electronic medical record using a standardized Lung-RADS reporting system. Supported in full by the James Comprehensive Cancer Center, all positive scans are reviewed by a board-certified pulmonologist who is expert in lung cancer early detection and nodule management. A certified nurse practitioner and oncology coordinator work to ensure efficient clinic flow and a volunteer to escort patients to and from their CT scan. All CT results are discussed with the patient and they receive a printed copy. In addition, a clinic letter is forwarded to the patient's primary care physician to ensure adequate follow-up of incidental findings.

All patients seen at the LCSC are counseled on smoking cessation and a formal referral is offered. If agreed upon by the patient, scheduling staff make the cessation appointment, in addition to their next screening appointment. A pamphlet describing OSU's Tobacco Dependency Clinic is offered to each patient, if they decline referral. The pamphlet describes the benefits of the smoking cessation clinic and how patients can schedule an appointment. Patients who enroll in the cessation program will be excluded from this study due to possible confounding factors and so as not to distract from the cessation process. The LCSC estimates that approximately 10% of patients enroll in the smoking cessation program following their LCSC appointment.

Should the patient be found to have lung nodules requiring additional management, they will be immediately referred to the Pulmonary Clinic and excluded from the study.

Recruitment fliers will be posted in other clinics (e.g. OSU East and CarePoint East), and on social media platforms (e.g. Facebook, Instagram, ResearchMatch (RM)). Eligibility criteria will be assessed by phone and consenting will occur in person at the LCSC or by email.

4.3 Participant Screening and Informed Consent

The Project Manager (Co-Investigator) will provide a brief overview of the purpose and procedures of the trial. If the potential subject is interested in participating, their contact information will be collected. Any personal health information will be kept confidential, and data will only be identified by subject numbers at all stages of the recruitment and informed consent process. When eligible candidates have met preliminary screening criteria (**Appendix 4: Eligibility Screener**), the Project Manager will initiate the informed consent process with participants. Once all questions have been addressed, participants will be given time to review and sign the consent if time allows, or if not, they can mail it back. After consent is obtained all study visits will be scheduled. However, if participants do not have access to email a hard copy will be mailed to them with a pre-stamped return envelope.

Under the supervision of the Screening Clinic Director (Co-Investigator), all participants will be consented and screened by the Project Manager at the OSUCCC-LCSC. Information about the study will be provided and any questions will be answered. Written informed consent will be obtained before any procedures are performed.

This includes the signature, date, and time consent was obtained, as indicated by the research subject. The witness signature will be signed where applicable.

Consented participants will be registered with the Clinical Trials Office (OnCore). All participants will be provided a copy of the signed consent form and HIPAA Authorization form. The original documents will be filed in a locked study cabinet in the PI's locked office. The decision not to partake in this study will not affect a participant's treatment at OSU. Information on participants undergoing the recruitment process but deciding not to partake in the study will be deleted immediately. Any subsequent changes to the informed consent process will be submitted to the OSU IRB for approval prior to activation.

4.4 Other participant considerations

4.4.1 Inclusion of Women and Minorities

Women will represent 50% of the study participants. Although there is limited evidence that women and men metabolize phytochemicals differently, we aim to investigate this, and, therefore, an equal number of women and men have been factored into our study design. In our capture area, approximately 17% of the population is defined as minority. In recent OSU-CCC clinical intervention studies, approximately 17% of the participants enrolled were of minority groups. No patient will be excluded from participating on the basis of race or ethnicity.

4.4.2 Inclusion of Children

Children do not qualify for CT screening and are, therefore, excluded from this study.

4.5 Registration

All patients will be registered and enrolled by the study personnel specifically assigned to this clinical trial. All patients will be required to sign an Informed Consent Form in addition to a HIPAA Form as it relates to this specific clinical trial. All patient information will be kept confidential in a REDCap database specific to this project.

5. Study Procedures

The study procedures include following a low polyphenol diet, consumption of 2 x 170 mL BRB nectar daily, returning all stool and urine samples to clinic, and attending four additional clinic visits with successful blood draws, nasal swabs and body measurements. The procedures are detailed successively below:

5.1 Screening and Instructional session (-4-0 weeks)

Care has been taken to ensure that consent flow and screening do not impede normal clinic operations. Standard clinic practice involves seeing Nurse Practitioners before administration of the chest computed tomography (CT) scan, after which patients normally wait approximately 30 minutes for their results. Following the CT scan patients will be led to a private exam room not used during the LCSC hours, provided by the Clinic Manager. During the wait time, the Project Manager will ask patients if they are interested in participating in this trial and will conduct the screener for eligibility (**Appendix 4: Eligibility Screener**). Interested patients will sign informed consent, and will be instructed on a "controlled phenolic diet" (**Appendix 1: Low Phenolic Diet Compliance**) that we have developed and tested for compliance. This checklist, which minimizes very high food sources of polyphenols, was found to be acceptable and achievable, whereas a polyphenol-free diet is overly restrictive and nearly impossible to achieve³⁹.

Next, participants will undergo a web-based interview to collect a Detailed History (**Appendix 5: Medical History Questionnaire**) on ethnicity, education, income, age, sex, race and medical status. They can either use ipads provided in the clinic or other computer offsite using an emailed link. Data will be stored in the RedCap database.

Participants will also be asked to complete a Food Frequency Questionnaire (FFQ) (**Appendix 2: Food Frequency Questionnaire**). Study participants will complete the Viocare electronic FFQ prior to the intervention (at the week 0 visit) which will be analyzed with corresponding software by the OSU Clinical Research Center (CRC) (part of the Center for Clinical and Translational Sciences, CCTS)⁴⁰. This FFQ estimates long-term dietary patterns and has been tested for validity and reliability in clinical and epidemiology studies. This data may be used by the individual projects to provide an overall description of the diet composition among smokers and nonsmokers and may be used as a covariate in estimating the effect of the BRB nectar on the microbiome, as diet affects both the microbiome and inflammation.

Participants will also be asked to complete a Physical Activity Questionnaire (**Appendix 10: International Physical Activity Questionnaire**). This is an internationally-vetted activity questionnaire that asks about the time spent being physically active in the last seven days. This form is seven questions long and takes an average of five minutes to complete. These data may be used as a covariate in estimating the effect of the BRB nectar on the microbiome, as physical activity also affects the microbiome.

5.2 Randomization

Prior to study start, participants will be randomized to the “Placebo First” (PF) or “BRB First” (BF) groups via the REDCap randomization module, stratified by sex and smoking status. A block randomization design will be used with a 50% probability for each females and males to be assigned to PF or BF, and a 50% probability of current smokers and former smokers to be assigned to BF or PF. The LCSC estimates 60% of their patient population to be male, which calculates to 8 Female PF, 7 Female BF, 14 Male PF and 13 Male BF. The LCSC estimates 60% of their population to be current smokers, which will follow the same patterns as above. We will not include an interaction term for the effects of BF or PF on outcomes by sex and smoking status. If differences between PF and BF are not observed after trial conclusion, thereby indicating that the washout period was of sufficient duration to prevent carryover effects, the BF or PF covariate will be dropped from all further analyses. Patients actively seeking to change their smoking status will be excluded (See 4.1.2. Exclusion Criteria). All calculations were performed with the randomizer package in R.

5.2 Study Visits (Visits 1-4, Weeks 0, 4, 6 and 10)

All study visits will occur at the OSU Clinical Research Center (2115 Davis Medical Research Center, 480 Medical Center Drive). Trained staff will facilitate the collection and measurement of data and samples described below. Parking vouchers will be provided to study participants.

5.3 Black Raspberry Nectar or Placebo (twice daily for Week 0-4 or 6-10)

Each participant will receive 170 mL placebo or BRB nectar to be consumed twice daily for each 4-week treatment period (see **Appendix 6: Beverage Instructions**). The BRB nectar is a novel and high quality, bioactive-rich delivery vehicle that enhances the exposure to desired phytochemicals. The placebo nectar has raspberry flavoring and red food grade coloring, routinely used in the US food industry, will be added to provide taste characteristics very similar to the BRB nectar. Patients will complete the beverage compliance form after they consume each product (see **Appendix 1: Low Phenolic Diet Compliance**).

5.4 Biological sample collection (Week 0, 4, 6, and 10)

Nasal swabs and blood samples will be collected by certified clinic staff in the CRC at Week 0, 4, 6, and 10 clinic visits. Urine and stool samples will be collected at home and returned to clinic during Week 0, 4, 6, and 10 visits. The samples collected are briefly described below (see Section 9.1 for complete methodological details of sampling).

5.4.1 Nasal brushings

Nasal epithelia will be collected with a Celletta™ brush cell collector with protective tip (GmbH, Germany) and directly deposited into RNeasy lysis solution.

5.4.3 Blood samples

Blood will be collected using standard hospital clinical laboratory techniques for blood sampling via venipuncture. CRC staff will collect 18 mL blood in 3x6mL tubes.

5.4.4 Urine samples

Participants will be provided pre-weighed 24-hour urine containers prior to their visit. The containers contain 2g boric acid as preservative to prevent bacterial growth. 24-hour urine will be collected at clinic visits and analyzed for phytochemicals that indicate compliance with the study protocol (**Appendix 7: 24-Hour Urine Collection Instructions**).

5.4.5 Stool samples

Patients will be given OMNI Gene GUT OMR-200 home sampling kits (DNAgenotek) and stool collecting tub at consented and each subsequent study visit and are to return samples at clinic for analysis of microbiome that includes instructions in 14 languages and an instructional video (for instructions in English, see **Appendix 8: Stool Collection Instructions**).

5.4.6 Body measurements

Patients will have their height recorded at Visit 1 (Week 0), and their weight, heart rate (wrist pulse) and blood pressure recorded at Visits 1-4 (Weeks 0, 4, 6, and 10). Certified CRC staff will perform the collection.

5.5 Washout Period

Each participant will undergo a two-week washout period between treatment regimens to conservatively control for possible carry-over effects in our crossover design. Patients are to maintain a low phenolic content diet throughout the duration of the study, including the washout period.

5.6 Other study considerations

5.6.1 Patient Compliance

We have developed a checklist of the most common food sources of polyphenols based on the USDA Flavonoid Database ⁴¹. The checklist will be completed each day to document the consumption of any foods that are high in polyphenols (see **Appendix 1: Low Phenolic Diet Compliance**). The diary will also be used to document time of consumption of the beverages. All containers (boxes) of unconsumed beverage will also be returned at the end of the study visit. These boxes will be used to estimate compliance to the BRB nectar or placebo and chemical analysis of BRB compounds.

5.6.2 Patient Remuneration

Each patient will be given a \$50 Amazon or Walmart gift card for each of their four visits on this trial (0, 4, 6 and 10 weeks), for \$200 total per patient. We expect this remuneration scheme to increase patient retention, which will significantly affect the trial duration. As described previously, our sample size calculation (42 total patients) assumes a 0.6 retention rate. However, the power calculations, owing to our statistically-efficient crossover design, estimate 80% power to detect a difference in a cohort of size 26 (assuming full-compliance with all sample collections). Assuming a 10% accrual rate (~ 1 patient per 4/hr clinic) this represents reaching full accrual 8 weeks ahead of schedule (16 patients / 2 clinics per week). Distribution and record-keeping of gift cards will be managed through the CRC.

5.6.3 Early Termination from Study

Participants who are terminated due to noncompliance or multiple missed appointments or who voluntarily withdraw from study before completion of the full study will be replaced. Those participants who have begun an intervention period (week 0 or 6) and wish to withdraw prior to the end of the intervention (week 4 or 10) will be asked to return stool and urine collections to the HRLC, but will not be asked to provide a blood or nasal swab sample and will not be compensated. Those participants who withdraw during the washout period will be removed without the additional collection of samples or

distribution of materials (**See Study Calendar: Study Supply Distribution**).

Participants who experience a grade I or II toxicity attributable to the treatment product that want to discontinue use, will have the option to continue all other study activities and re-challenge at a later date. Those participants who begin a medication or substance that precludes eligibility will also have the opportunity to continue as if they were still on-study. As described above and in section 3.6, participants under either circumstance will be replaced and their data will be used in an intention-to-treat analysis.

6.0 Study Calendar

<i>Phase</i>	Screening	Treatment: BRB/PLACEBO		Washout	Crossover: PLACEBO/BRB	
<i>Timing</i>	<i>Week -4 - 0</i>	<i>Week 0</i>	<i>Week 4</i>	<i>Week 4-6</i>	<i>Week 6</i>	<i>Week 10</i>
Informed consent	X					
Detailed history interview	X					
Food frequency Questionnaire (FFQ)	X					
Physical Activity Questionnaire	X					
Patient education ¹	X					
Randomization ²	X					
Study supply distribution						
Daily diary	X					
BRB/Placebo Beverage		X			X	
24-hour urine container	X	X			X	
Stool sample kit	X	X			X	
Biological sampling						
Nasal brushings		X	X		X	X
Blood collection		X	X		X	X
Urine		X	X		X	X
Stool		X	X		X	X
Body Measurements		X	X		X	X
Study Treatment: BRB First Group (N=21)						
BRB nectar		X	X			
Placebo beverage					X	X
Study Treatment: Placebo First Group (N=21)						
BRB nectar					X	X
Placebo beverage		X	X			

7.0 Potential Risks

We divide risks into 4 categories: (1) Noninvasive testing, (2) Biological sample procurement, (3) Study agent, and (4) Confidentiality. Each will be reviewed below:

(1) Noninvasive testing. The non-invasive procedures, including completion of questionnaires (health and lifestyle questionnaire, Food Frequency Questionnaire, and sensory evaluation) and anthropometric measurements, represent minimal to no risk, and all are standard in a clinical care setting.

(2) Biological sample procurement risks. Blood collection, nasal swabs, 24-hour urine collection, and stool collection are the only procedures that will be completed with this study. Participants will be required to present themselves to the Blood Draw Clinic (BDC) at Martha Morehouse, downstairs from their standard clinic appointments, for scheduled appointments to provide blood and nasal swab samples and to collect urine and stool samples at home and present them to the clinic. Any risk associated with these procedures is as follows:

2.1 Urine:

Urine collection is low risk and without discomfort.

2.2 Stool:

Stool collection is low risk and without discomfort. Nitrile gloves will be included in each stool collection kit and instructions provided for how to remove gloves without exposing hands to glove outer surfaces. In addition, patients will be instructed to thoroughly wash hands after collection, as described in the stool collection instructions sheet (See Appendix).

2.3 Nasal brushings:

A thorough nasal swab entails approximately 10 seconds of swabbing the nasal cavity. This can result in minimal irritation, which resolves in 2 days.

2.4 Blood:

We will collect approximately 18 ml of blood, and all blood will be collected by the CRC staff. The volume of blood we will be collecting should not increase the risk of anemia.

2.5 Body Measurements.

Collection of height, weight, blood pressure and heart rate (wrist pulse) is low risk.

(3) Potential risks of study agent: The study is a nutritional study of a novel BRB food product which will be tested for its impact on biomarkers relevant to its action as a modulator of carcinogenesis. The following are detailed during the informed consent processes and summarized here.

There are no documented reports of BRB toxicity in the published literature other than among people who have an allergy to BRBs or a contaminant. The symptoms of a berry allergy include but are not limited to: tingling, swelling and itching of the mouth, throat and tongue or lips, a runny nose, watery eyes, sneezing, and dermatitis. Men and women with a known intolerance or allergy to BRBs or other berries will not be enrolled in this study. We will assess each patient for any adverse event using the Common Terminology Criteria for Adverse Events (version 5.0) from the National Cancer Institute and National Institutes of Health. Participants will be asked about any adverse events at each visit and during each phone call. Graded toxicity scores will be assigned to any patient who reports an adverse event which may be related to the study.

(4) Confidentiality: Participant confidentiality will be maintained per OSU and NIH requirements including HIPPA (Health Insurance Portability and Accountability Act of 1996 Privacy and Security Rules). Individuals will not be identified by name or by any other personal identifying information in laboratory records, reports, or publications resulting from this study. All data will be coded and only the principal investigators and study coordinators will have access to personal subject information. All databases are maintained on password protected computers and any paper documents will be kept in locked file cabinets accessible only to the PI and study coordinators. Thus, risk of disclosure of confidential medical or personal information will be virtually nonexistent.

7.1 Protection against risks

The eligibility criteria limit the study to a population of men and women in good health. The project team has procedures in place to maintain confidentiality, such as password protected computer access, locked file cabinets and rooms where records are kept, have received appropriate training, and are in full compliance with

HIPAA. Any grade 2 adverse events will be reported to the IRB and the Comprehensive Cancer Center/DSMB. If we see 2 similar grade three adverse events, we will stop the study, ask all men and women to discontinue the intervention, and report these events to the IRB and our Comprehensive Cancer Center.

7.2 Adverse Events

We will assess each patient for any adverse event using the Common Terminology Criteria for Adverse Events (version 5.0) from the National Cancer Institute and National Institutes of Health. Participants will be asked about any adverse events at each visit and via phone call 2 weeks into each intervention period, and given space to document any events on a daily compliance form (See **Appendix 9: Daily Beverage Consumption Compliance**). Graded toxicity scores will be assigned to any patient who reports an adverse event which may be related to the study.

If we see 2 similar grade three adverse events, we will stop the study and ask all men and women to discontinue the intervention. Discoloration of teeth may occur with the BRB nectar, but this discoloration should be completely reversible with regular and thorough oral hygiene.

Serious Adverse Events: The likelihood of serious adverse events is nearly nonexistent. The compounds in the foods provided are commonly found in a general, varied diet. However, in the unlikely event of any life-threatening reactions, all fatal events while on study or within 30 days of treatment, and the first occurrence of any previously unknown toxicity of any grade will be reported as serious adverse events. The Study Coordinator will report these reactions within 24 hours to the National Cancer Institute Investigational Drug Branch and the IRB within 10 working days. The principal investigator and the attending physician in collaboration with the NCI, if necessary, will decide if a toxicity is most likely drug-related.

7.2.1 Dose Modifications

Any participant who reports intolerance to the study product of grade 2 toxicity will be permitted to reduce the daily dose to half the study product (85 mL 2 x per day) for the duration of the study. A grade 3 toxicity or a request by the patient to stop consuming either beverage will result in suspension from the study.

7.2.2 Suspension of participation

Participants may voluntarily withdraw from study protocol at any time without impact upon their care or treatment. In such an event, if a patient participant enrollee directs, any relevant biosamples will be destroyed (if patients have previously given permission for their identifiers to be linked with their samples after which such samples can be traced and destroyed). Otherwise, the data available on the patients who drop out will be used in an intention-to-treat analysis in order to minimize potential missing data bias. Any subject who decides to leave the study will be questioned to determine the reason for discontinuation. Participants are not obligated, however, to provide a specific rationale for their decision, though a reasonable effort will be made to determine the cause which will be documented. If patients consuming either beverage experience toxicity attributable to use of the treatment product (attributed as probable or definitely related), and it is the opinion of the physician that patient(s) should discontinue the product use, such participants will be instructed to discontinue administration, and they will be removed from the protocol. The Investigators may also decide to suspend a participant's participation if the participant refuses to consume the intervention product, begins certain medications or substances after trial enrollment (such as antibiotics), or develops a grade 3 toxicity. As above, discontinuation initiated by the principal investigator will not impact the patient's care or treatment and all participants will receive compensation prorated to equal their participation in the study.

7.2.2.1 Non-compliance Definition:

Participants will be considered non-compliant, and their participation will be terminated under either of the following circumstances:

- The subject fails to consume study products (less than 65% consumption of the beverages).

- The subject is absent from more than two rescheduled appointments or if follow-up contact is lost (no response to email and telephone messages). Attempts to contact the subject will be documented as a note to the subject's file.

8. Study Agent Information

8.1 Product Description

A novel BRB delivery system, the BRB nectar was developed at OSU under the direction of Co-Investigator. The approach utilized state-of-the-art food technology to develop a novel and high quality, bioactive-rich delivery vehicle that enhances the exposure of oral mucosa to desired phytochemicals. The placebo nectar will have raspberry flavoring and red food grade coloring, routinely used in the US food industry, will be added to provide taste characteristics very similar to the BRB nectar.

8.2 Toxicology

There are no documented reports of BRB toxicity in the published literature other than among people who have an allergy to BRBs or a contaminant. The symptoms of a berry allergy include but are not limited to: tingling, swelling and itching of the mouth, throat and tongue or lips, a runny nose, watery eyes, sneezing and dermatitis. Men and women with a known intolerance or allergy to pectin, BRB, other berries, or products made from berries will not be enrolled in this study.

8.3 Solution Preparation

8.3.1 Preparation of BRB Freeze Dried Powder:

The whole berries are shipped frozen overnight from the Stokes Farm in Wilmington, OH, to Van Drunen Farms in Momence, Illinois, where they are immediately freeze-dried in a commercial grade food lyophilizer. During the freeze-drying process, approximately 3,500 pounds of berries are dried in a Virtis sublimator under vacuum. Since berries contain approximately 85%-90% water, 1 gram of the lyophilized product is equivalent to 10 grams of wet black raspberries. The dried berries are then ground into a powder, stored at -20 °C, and shipped frozen to OSU. Since the lyophilized (freeze-dried) BRB products are not mechanically or chemically sterilized, we store them in 20 lb aliquots in a single -20 °C freezer. Although a loss of electrical power is not critical for the preservation of the powdered berries, all freezers in Parker Food Science are alarmed with battery backup.

8.3.2 Nectar processing and production:

We have utilized state-of-the-art food technology to develop a novel, well-characterized, high quality, and bioactive-rich delivery vehicle that enhances the exposure of oral mucosa to desired phytochemicals. The BRB nectar contains 5.6% BRB powder, 3% sugar, 1% corn syrup, 89.8% water and 0.5% pectin. In order to extend the quality and safety of the nectar product, the nectar will be sterilized and aseptically packaged. In the preparation, pectin, sugar, corn syrup and water will be first mixed held at 25°C in a holding tank held at ambient temperature. BRB powder will be added to this mixture and will be sterilized by a flash pasteurization process (commonly known as high-temperature short-time or HTST) at 165 °C for 23 sec (adapted from Silva and Gibbs for inactivation of spoilage enzymes and molds(Meyer et al., 2008)), cooled to 25 °C, and filled into 185 mL polyethylene cardboard boxes under sterile conditions which include the use of an industrial scale aseptic filling system. Nectars have a one-year shelf-life when held at room temperature which can be extended to a second year with storage at -20°C. Therefore, all nectars will be stored at -20°C until distribution to subjects. This optimized product was used in an initial phase II trial (see Preliminary Data for details) and demonstrates its ability to deliver BRB phytochemicals to the oral mucosa with high adherence/acceptability from study participants. Total kcal in the 10 gm a 185 mL boxes of nectar is 92.5 Kcal (calories from BRB are 52.5 Kcal).

8.3.3 Processing of Placebo Beverage:

The placebo formulation will contain the same pre-mix and will undergo the same processing as the BRB-based nectar, except that no BRB will be added. Raspberry flavoring, purple food grade coloring, and citric acid (Kool-Aid, Kraft-Heinz Corp, Northfield, IL) will be added to the placebo beverage in place of the BRB. This will provide the organoleptic qualities that are similar to the BRB nectar described above. Total kcal in the placebo (185 mL) nectar is 42.5 Kcal.

8.4 Storage Requirements

8.4.1 Lyophilized BRB powder:

The research berries are mechanically harvested as a single lot, immediately washed, and frozen at -20 °C on the farm property within 2 hours of harvesting. OSU investigators have shown that 26 micronutrients remain within 10%-20% of the original measurements for at least 2 years storage when freeze dried at -20 °C. We will obtain BRB over the first 3 consecutive growing seasons. The micronutrient content of berries harvested in different years is quite reproducible in which most measured components usually vary by <20%.

8.4.2 Finished Beverages:

Once the beverages have been pasteurized, they will be used to fill coded, shelf-stable, aseptic containers. The boxes will be coded with a five digit box identifier and dated with the year and the three digit Julian calendar day in which the beverages were processed. Only study staff will be aware of the identity of the nectar, and Principal Investigator (PI) will supervise the randomization. Participants will be able to store their boxes in a dry, cool space for all 4 weeks that they are consuming the product. The beverage will be distributed at two time points in the study to correspond to beginning the BRB and Placebo treatments, respectively (see Study Calendar in Section 6).

8.5 Stability

Ensuring the consistency and stability of bioactives in the food product employed in the clinical trial is essential for proper testing of outcomes traceable to BRB phytochemicals. Samples will be evaluated at each step of the process, during which they will be: (i) frozen harvested, (ii) freeze-dried immediately upon arrival from processor, (iii) freeze-dried prior to nectar preparation, (iv) made into fresh nectar, (v) stored at OSU for 3 months as processed nectar samples, and (vi) obtained as boxes of nectar from each participant after 6 weeks of storage under home conditions. Analysis will be completed by the OSU-Comprehensive Cancer Center Phytochemical and Nutrient Analytic Core as described in facilities and resources section and below in analytic methods. In brief, analysis of phytochemicals will employ HPLC with photodiode array (PDA). BRB polyphenols representative of each class will be determined for quality control purposes. These species will include cyanidin rutinoside for anthocyanins, quercetin rutinoside for quercetin glycosides, ellagic acid, and sanguiin H-6 for the ellagitannins. Polyphenols will be extracted from nectar samples using previously published methods and quantified using high performance liquid chromatography (HPLC). For ellagitannin determination we have applied a recently published method. Based upon our development studies and prior publications with BRB consistency of < +/-10% is expected across the entire study.

8.6 Route of Administration

Participants enrolled in both arms will be instructed to consume one 170 mL container of their assigned beverage per day. Participants will be encouraged to consume their beverage at a similar time each day.

8.7 Safe Handling

There are generally no specific requirements for the handling of the BRB nectar or placebo other than accepted good hygiene practices, as with any food product. The finished beverages will be packaged in 250 mL aseptic containers and distributed to the participants will be instructed to keep the boxes in a dry, cool place in their homes; they do not need to be refrigerated.

8.8 Supplier

It is crucial that the product used is chemically and biologically uniform. To ensure a consistent agent, the BRB (*Rubus occidentalis*) will be obtained from the same farm in southern Ohio to eliminate area and growing differences (Stokes Berry Farm, Wilmington, OH; www.stokesberryfarm.com). The Stokes' Farm is the largest producer of black raspberries outside of California and Oregon. Dale Stokes and his son, Mark, are leading authorities on varietal and horticultural techniques and have been involved with black raspberry research conducted by many investigators at OSU for over a decade (6,000 lb/yr provided to OSU). In collaboration with the University of Maryland, the Stokes Farm has the most advanced black raspberry breeding evaluation and micropropagation program in the nation. OSU investigators have shown that 26 micronutrients remain within 10%-20% of the original measurements for at least 2 years storage when freeze dried at -20° C. We will obtain BRB over the first 3 consecutive growing seasons. The micronutrient content of berries harvested in different years is quite reproducible in which most measured components usually vary by <20%.

As the BRB nectar and placebo beverage are primarily food-based products and contain ingredients that are GRAS (generally regarded as safe) approved, an IND application is not necessary for this protocol. The beverages will be manufactured at Island Aseptics in Byesville, OH (<http://www.islandoasis.com/company/manufacturing-capabilities.html>). This facility is Global Food Safety plants accredited by BRC (British Retail Consortium) standards and inspected annually by a third-party audit. The BRB nectar will be prepared using GMP (good manufacturing practices) under the supervision of Co-Investigator.

9. Biologic Sample and Health Information Collection

9.1 Biologic Sampling

Methods for collection of biological samples are described below.

9.1.1 Blood

Blood will be collected by researchers or nursing staff certified in phlebotomy using standard hospital clinical laboratory techniques for blood sampling via venipuncture. Whole blood will be collected in three 6 ml glass tubes. Blood will be immediately processed for the following analyses, or stored in a freezer at -80 °C in the lab of PI until processing in batch. The volume of blood sampling is modest and should not result in or contribute to anemia unless some additional undiagnosed condition exists.

9.1.1.1 Serum samples

Serum will be stored for validation of changes observed in inflammatory genes. For example, CRP, IL6 and TNF-alpha levels will be analyzed using a Behring Nephelometer Analyzer System (Behring Diagnostics, Inc., Somerville, NJ, USA).

9.1.1.2 Immunostaining and FACS sorting:

NKT cells will be stained and sorted to determine their abundances pre and post intervention. Appropriate cell surface antibodies (CD1d-loaded tetramers with α -galactosylceramide (GalCer), β -sulfatide and other glycolipids, V α 24, V β 11, CD45, PD-1, PD-L1, IL-10, TGF- β , IL-4, IL-13, IFN- γ and other antibodies targeting novel cell surface markers and intracellular cytokines) will be used to stain the cells that will be sorted from peripheral blood mononuclear cells (PBMCs) isolated from blood collected in Heparin tubes.

9.1.1.3 Cytokine analyses:

Intracellular cytokine production within immune cell subsets will be analyzed by the cytokine bead array and single-cell polyfunctional strength assay (i.e., IsoCode, Isoplexis). Both technologies allow for simultaneous detection of up to 40 cytokines from limited material.

9.1.1.4 Time-of-flight mass cytometry (CyTOF):

Relative abundances and types of immune cells will be estimated by CyTOF of PBMCs, which allows high-dimensional detection and analysis of both surface and intracellular proteins, such as Foxp3, Ki-67, CD45RA, CTLA-4, PD-1, TGF- β and IL-10, NK1.1, IL-13 and others.

9.1.1.5 T cell receptor (TCR) immunosequencing (immunoSEQ):

Type I and type II NKT cells can be distinguished by the usage of the α and β chains of their T cell receptors (TCRs), while in type I NKT cells the V α 24 gene segment pairs almost exclusively with the V β 11 gene segment, type II NKT cells show a more diverse TCR repertoire, with the most frequently found rearrangements being V α 1 and V α 3 paired with V β 8.1/V β 8.3⁴². TCR diversity and repertoires of the α and β chains of the TCR will be analyzed via TCR immunoSEQ (Adaptive Biotechnologies) from DNA isolated from PBMCs previously sorted by flow cytometry (see 9.1.1.2: Immunostaining and FACS sorting).

9.1.1.6. NKT cell culture:

We will sort V α 24+V β 11+ NKT cells via flow cytometry and expand them by culturing PBMCs with autologous dendritic cells (DCs) in the presence of 100 ng ml⁻¹ α -GalCer and 10 ng ml⁻¹ IL-7 for 10 days, followed by 40 U ml⁻¹ IL-2. On day 14, V α 24/V β 11 double-positive cells will be sorted via flow cytometry and kept in culture with the addition of IL-7 weekly and IL-2 every 3–4 days. Cultured NKT cells will be used for in vitro experiments. If any remain, they will be expanded for up to 14 days after the beginning of the culture, then cryopreserved for future experiments/analyses. Any cryo-preserved samples will be destroyed within 1 year following publication.

9.1.1.7 Regulatory T cell (Treg) suppression assay:

We will also co-culture autologous regulatory T cells (CD4+ CD25+ FoxP3+) with CFSE-stained effector T cells with or without expanded NKT cells for several days and subsequently detect their proliferation via flow cytometry.

9.1.1.8 Immune senescence:

CD3+ T-cells will be isolated using RosetteSep or similar commercial reagents for negative selection. Nucleic acid will be extracted from purified PBTL or PBMCs using a column-based commercial nucleic acid purification kit. Changes in T-cell epigenetics will be analyzed by Zymo Research Corp., TruDiagnostic, or an equivalent vendor to characterize T-cell Horvath clock/DNAge and gain further insight into immune function. No identifying genomic information is obtained in these assays.

9.1.2 Nasal brushings

During blood draw appointments at the BDC participants' nasal cavities will be swabbed by certified clinic staff. Nasal epithelia will be collected with a Celletta™ brush cell collector with protective tip (GmbH, Germany) and directly deposited into RNAprotect Cell Reagent solution. Total RNA will be purified using the All-in-One purification kit (Norgen Biotek) and its integrity assayed by Agilent bioanalyzer (Agilent Technologies, Santa Clara, CA). Ribosomal depletion will be performed with the RiboGone-Mammalian kit (Clontech Cat. Nos. 634846 & 634847) and cDNA created with the SMARTer Stranded RNA-Seq Kit (Cat. Nos. 634836). Samples will be sequenced using an Illumina HiSeq 4000 with 2x125 bp reads, with an average of 47.5 million reads per sample. Processing will use the exceRpt pipeline; which PI has modified for long-read transcripts [42]. Briefly, transcripts will be aligned to the human reference (hg38) using STAR [43] and quantified using RSEM [44]. Unaligned reads will then be processed as in the case of stool samples, via direct alignment to genomes and then alternative methods.

9.1.3 Urine:

Participants will be provided pre-weighed 24-hour urine containers prior to their urine collection. The containers contain 2 g boric acid as preservative to prevent bacterial growth. Patients will be asked to collect urine 24 hours before each clinic visit (weeks 0, 4, 6 and 10). Complete 24-hour urine pools will be collected at the BDC appointment and volume will be recorded. The methods are briefly described in the educational material provided to study participants (Appendix 7: 24-Hour Urine Sheet). After collection, the containers will be re-weighed, and five 5 mL of tubes will contain sufficient formic acid to achieve a final formic acid percentage of 5%, and another five 5 mL tubes will have no acid. Both sets will be stored at -80°C for analysis.

9.1.4 Stool:

Participants will be given one OMNI Gene GUT OMR-200 home sampling kits (DNAgenotek) or stool collecting tub in each distribution set (See Study Calendar) and asked to collect samples within 24 hours of each BDC visit (See Appendix 8: Stool Collection Instructions). This method has been validated by large scale, NIH funded studies⁴³. Briefly, participants will defecate onto a narrow paper sheet placed onto the toilet bowl. A small, disposable spatula will be used to transfer ~ 1g stool into the cap of the OMR-200. Screwing on the cap releases nucleic acid stabilizer solution and a ball bearing disperses and the stool. The sample is then stable at room temperature for several weeks. Nucleic acid purification will be performed by PI using a QIAGEN PowerFecal kit and libraries created using the Nextera DNA Flex Library Prep Kit (Illumina, USA). Microbes will be analyzed by shallow mWGS sequencing (>2 million reads/sample) on a HiSeq 4000 (Illumina, San Diego, CA, USA) or similar (e.g. NovaSeq 6000) at the Nationwide Children's/OSUCCC Biomedical Genomics Core, or other microbiome sequencing vendor (OneCodex, San Francisco, CA), with higher-depth sequencing in a subset of samples where thorough strain characterization is warranted. PI will perform all data cleaning, processing and analysis. Briefly, reads will be filtered for quality and human DNA (estimated 10%). Microbe reads will be aligned to curated genomes (Ensembl).

9.2 Personal Health Information Collection and Non-invasive Testing

Data collected on research case forms will be kept confidential in a locked filing cabinet in PI's private office in the Biomedical Research Tower, room 0482. Data will be converted to electronic formats when appropriate and stored behind a firewall in the REDCap web-based data storage system. This system has been approved for storage of all sensitive data including PHI.

9.2.1 Personal Data:

We will retain data used during screening for all enrolled persons. This includes personal health information, demographic data, smoking history, and alcohol use.

9.2.2 Diet Data:

At the Week 0 visit, participants will complete an electronic Food Frequency Questionnaire (Viocare FFQ). Participants' data will be coded and used to describe the macronutrient and micronutrient average consumption of this population. It will also be used to estimate habitual berry consumption.

9.2.3 Anthropometric and Vital Sign Data:

Weight, height, blood pressure, heart rate, temperature, and respiratory rate for all participants will be recorded in the LCSC and extracted to the REDCap database.

10. Data Safety Monitoring Plan

The data and safety monitoring plan will involve the continuous evaluation of safety, data quality and data timeliness. Investigators will conduct continuous review of data and patient safety at regular disease group meetings (thoracic, monthly; LCSC, quarterly) and the discussion will be documented in the minutes. The Medical Oncologists (PIs) will review toxicities and responses of the trial where applicable at these disease group meetings and determine if the risk/benefit ratio of the trial changes. Frequency and severity of adverse events will be reviewed by the Medical Oncologist (PI) and Carbone (PI) and compared to what is known about the agent from other sources; including published literature and scientific meetings to determine if the trial should be terminated before completion. Serious adverse events and responses will also be reviewed by the OSUCCC Data and Safety Monitoring Committee (DSMC). All reportable Serious Adverse Events (SAE) will also be reported to the IRB of record as per the policies of the IRB.

10.1 Protection of Subject Confidentiality:

Strict patient confidentiality of records will be maintained. Random 3-digit numbers will be assigned to each subject within the REDCap database only the key will be used to identify participants' samples and files.

Individuals will not be identified by name or by any other personal identifying information in laboratory records, reports, or publications resulting from this study. Any paper records containing Private Health Information will be stored in locked file cabinet in PI's locked, private office (BRT 0482). All team members have received appropriate training, in full compliance with HIPAA. Only the investigators and study coordinators will have access to any subject data, which will be managed via access to the REDCap database. Personal identifiers will be kept until 2 years after the publication of results and deleted thereafter.

11. Statistical Analysis

All statistical analyses will be performed with support from the Biostatistics Co-Investigator.

11.1 Primary Feasibility Endpoint

The primary aim of the study is to evaluate the feasibility of collecting longitudinal, repeated biospecimen collection at OSUCCC-LCSC. We will recruit up to 42 eligible patients for this pilot study. The study is planned to determine whether dietary intervention and biospecimen collection is acceptable in this patient population. To evaluate feasibility, we will determine recruitment and compliance rates. Recruitment rates will be defined as the proportion of screened accrued relative to those approached in OSUCCC-LCSC. A non-enrollment survey will be used to capture why participants choose not to enroll.

Successful recruitment rates will be defined as $\geq 10\%$ of participants approached agree to participate. Study adherence will be defined as the percent of the biospecimens collected at the end of the study period, with a compliance goal of 60%. This 40% loss of biospecimens is accounted for in the power calculations. We will track the monthly collection of data and document reasons for missing any scheduled collection dates. The proportions of missing information at each time point will be calculated. Descriptive statistics will examine the distribution of all patient and treatment characteristics, overall and by patient and disease characteristics. This study is powered to reject acceptability of these measures if the follow-up rate is less than 40% (10% one-sided type one error) and deemed feasible if the follow-up rate is 60% or more (90% power). If 26 of the 42 patients complete the study, we will consider this method to have acceptable compliance for further study. The recruitment rates and compliance will be used in combination with the variance of the biospecimen data in power calculations to estimate the sample size needed for the planned R01 application.

11.2 Primary Therapeutic Endpoints

The primary therapeutic endpoints of this trial are (1) a decrease in nasal brushings inflammatory markers by RNAseq, and (2) changes in stool microbe relative abundances by mWGS. The paired-sample design of this crossover trial facilitates analyses that compare the treatment and placebo effects for each individual, thereby reducing the variance of the data. This facilitates 80% power to detect 1.8-fold changes in nasal-brushings RNAseq genes or stool mWGS microbes at the expected coefficient of variation (0.3) after correcting for 200 tests (Figure 2). Procedures for each of the primary outcomes and tests follow.

We will be monitoring various biologic responses to the BRB nectar. These endpoints are described here: samples of urine, stool, and blood will be analyzed for BRB phytochemicals (ellagitannins, ellagic acid, quercetin glycosides, anthocyanins) and their metabolites (urolithins, quercetin glucuronides/sulfates, methylated anthocyanins) using HPLC-mass spectrometry (LCMS). Metabolites in extracts of biosamples are resolved chromatographically and quantified by photodiode array detector and/or mass spectrometer via ion intensity with the selectivity of MS/MS transitions in targeted analyses. Authentic standards where available are used to calibrate response. For metabolomics experiments, urines are diluted and injected for LCMS directly while proteins are precipitated from bloods and before analyzing supernatants. Full scan MS is utilized with accurate mass and high resolution to detect and identify known and novel metabolites and compare levels between samples. Bioinformatics software is applied to reveal compounds unique to certain samples as well as fold changes for metabolites common between samples. Public databases are screened to find tentative identifications of unknown metabolites.

11.2.1 A significant difference in inflammatory genes between treatment and placebo (45 tests).

RSEM- calculated Transcripts per Million (TPM) values at treatment endpoint (week 4, 10) will be compared to baseline values (week 0, 6) using a Wald test statistic in DESeq2 in R⁴⁴, filtering to the 40 genes with the largest change in expression and including known pro-inflammatory genes such as CRP, IL6, and TNF- α . In addition, genes will be mapped onto KEGG pathways⁴⁵ and the differential abundance of pro- inflammatory KEGG pathways (e.g. TNF signaling pathway, NF- κ B signaling) will be compared between the two groups (student's T-test) using the package Pathview in R (~5 tests)⁴⁶.

11.2.2 A significant difference in microbe relative abundances between treatment and placebo (45 tests).

Microbe relative abundances will be compared between treatment and placebo as described for the gene expression, but with additional filtering to accommodate the sparseness of the microbiome data matrix. Specifically, microbes will be compared that are the most abundant, as well as being present in greater than 20% of the samples. In addition, specific taxa with literature precedence for a relationship to cancer or inflammation will be tested, including *A. muciniphila* and the genus *Clostridium*. Global differences between treatment and placebo will be measured by the alpha diversity metric Simpson's Index⁴⁷, the beta diversity metric weighted Unifrac⁴⁸, and by clustering with Non-Metric Multidimensional Scaling⁴⁹ with the significance of clusters defined by the multi-response permutation procedure⁵⁰, calculated in the vegan package in R⁵¹.

11.2.3 A significant correlation between microbes and inflammatory genes (20 tests).

Non-linear linking will be performed, which we have demonstrated to be more powerful than linear correlation methods in the context of noisy, sparse data⁵². Briefly, the 100 most highly correlated gene/microbe pairs and 100 least correlated (Pearson's correlation) will be used as training set for a random forest classifier. The features used by the classifier are topic models defined separately for the gene set and the microbe set using Latent Dirichlet Allocation (lda package in R⁵³). The classifier then defines gene-microbe linkages that can be organized by GO-term enrichment (topGO package in R⁵⁴). We've shown this LDA-based non-linear abstraction successfully reduces the noise in the dataset and results in stronger biological inference⁵². The result of this analysis will be clusters of gene-microbe pairs that change significantly with BRB nectar treatment and offers stronger hypotheses for the causal link between microbe and expression changes.

11.2.4 A significant difference in microbial metabolic pathway abundances between treatment and placebo (20 tests).

Much like the analysis described above regarding the enrichment of human pro-inflammatory pathways (1.1), microbe metabolic pathway enrichment or depletion will be evaluated. Microbe RNAseq reads will be directly aligned to metabolic pathways, independent of the microbial genome of origin. This method has the advantage of being less sensitive to biases introduced by incomplete microbial genome databases by using Hidden Markov Models (HMMs) for conserved gene signatures rather than more stringent direct alignment to genomes. We will align metagenomic reads to the FOAM database⁵⁵ from which HMMs have been made for microbial metabolic genes from KEGG orthologs, and map HMM alignments onto functional pathways. Total reads representing pathways of interest, such as short-chain fatty acid production will be compared between nectar and placebo treatments.

12. References

1. National Cancer Institute. Non-Small Cell Lung Cancer Treatment (PDQ®)–Health Professional Version. 2018.
2. Dickson RP, Huffnagle GB. The Lung Microbiome: New Principles for Respiratory Bacteriology in Health and Disease. *PLOS Pathogens*. 2015 Jul 9;11(7):e1004923.
3. Dickson RP, Martinez FJ, Huffnagle GB. The role of the microbiome in exacerbations of chronic lung diseases. *The Lancet*. 2014;384(9944):691–702.
4. Liang HY, Li XL, Yu XS, Guan P, Yin ZH, He QC, Zhou BS. Facts and fiction of the relationship between preexisting tuberculosis and lung cancer risk: a systematic review. *Int J Cancer*. 2009 Dec 15;125(12):2936–44.
5. Pallis AG, Syrigos KN. Lung cancer in never smokers: disease characteristics and risk factors. *Crit Rev Oncol Hematol*. 2013 Dec;88(3):494–503.
6. Schirmer M, Smeekeens SP, Vlamakis H, Jaeger M, Oosting M, Franzosa EA, ter Horst R, Jansen T, Jacobs L, Bonder MJ, Kurilshikov A, Fu J, Joosten LAB, Zhernakova A, Huttenhower C, Wijmenga C, Netea MG, Xavier RJ. Linking the Human Gut Microbiome to Inflammatory Cytokine Production Capacity. *Cell*. 2016 Nov 3;167(4):1125–1136.e8. PMID: PMC5131922
7. Shiels MS, Pfeiffer RM, Hildesheim A, Engels EA, Kemp TJ, Park J-H, Katki HA, Koshiol J, Shelton G, Caporaso NE, Pinto LA, Chaturvedi AK. Circulating Inflammation Markers and Prospective Risk for Lung Cancer. *J Natl Cancer Inst*. 2013 Dec 18;105(24):1871–1880. PMID: PMC3888091
8. Liu HX, Tao LL, Zhang J, Zhu YG, Zheng Y, Liu D, Zhou M, Ke H, Shi MM, Qu JM. Difference of lower airway microbiome in bilateral protected specimen brush between lung cancer patients with unilateral lobar masses and control subjects. *Int J Cancer*. 2018 Feb 15;142(4):769–778.
9. Yu G, Gail MH, Consonni D, Carugno M, Humphrys M, Pesatori AC, Caporaso NE, Goedert JJ, Ravel J, Landi MT. Characterizing human lung tissue microbiota and its relationship to epidemiological and clinical features. *Genome Biol*. 2016 Jul 28;17(1):163. PMID: PMC4964003
10. Routy B, Le Chatelier E, Derosa L, Duong CP, Alou MT, Daillère R, Fluckiger A, Messaoudene M, Rauber C, Roberti MP. Gut microbiome influences efficacy of PD-1–based immunotherapy against epithelial tumors. *Science*. 2018;359(6371):91–97.
11. Matson V, Fessler J, Bao R, Chongsawat T, Zha Y, Alegre M-L, Luke JJ, Gajewski TF. The commensal microbiome is associated with anti-PD-1 efficacy in metastatic melanoma patients. *Science*. 2018 Jan 5;359(6371):104–108. PMID: 29302014
12. Gopalakrishnan V, Spencer CN, Nezi L, Reuben A, Andrews MC, Karpinets TV, Prieto PA, Vicente D, Hoffman K, Wei SC, Cogdill AP, Zhao L, Hudgens CW, Hutchinson DS, Manzo T, Macedo MP de, Cotechini T, Kumar T, Chen WS, Reddy SM, Sloane RS, Galloway-Pena J, Jiang H, Chen PL, Shpall EJ, Rezvani K, Alousi AM, Chemaly RF, Shelburne S, Vence LM, Okhuysen PC, Jensen VB, Swennes AG, McAllister F, Sanchez EMR, Zhang Y, Chatelier EL, Zitvogel L, Pons N, Austin-Breneman JL, Haydu LE, Burton EM, Gardner JM, Sirmans E, Hu J, Lazar AJ, Tsujikawa T, Diab A, Tawbi H, Glitza IC, Hwu WJ, Patel SP, Woodman SE, Amaria RN, Davies MA, Gershengwald JE, Hwu P, Lee JE, Zhang J, Coussens LM, Cooper ZA, Futreal PA, Daniel CR, Ajami NJ, Petrosino JF, Tetzlaff MT, Sharma P, Allison JP, Jenq RR, Wargo JA. Gut microbiome modulates response to anti-PD-1 immunotherapy in melanoma patients. *Science*. 2018 Jan 5;359(6371):97–103. PMID: 29097493
13. Berzofsky JA, Olkhanud PB, Terabe M. Chapter Twenty - Complementary approaches to study NKT cells

in cancer. In: Galluzzi L, Rudqvist N-P, editors. *Methods in Enzymology* [Internet]. Academic Press; 2020 [cited 2020 Apr 30]. p. 371–389. Available from: <http://www.sciencedirect.com/science/article/pii/S0076687919303556>

14. Kepert I, Fonseca J, Müller C, Milger K, Hochwind K, Kostic M, Fedoseeva M, Ohnmacht C, Dehmel S, Nathan P. D-tryptophan from probiotic bacteria influences the gut microbiome and allergic airway disease. *Journal of Allergy and Clinical Immunology*. 2017;139(5):1525–1535.
15. Kelly CR, Kim AM, Laine L, Wu GD. The AGA's fecal microbiota transplantation national registry: an important step toward understanding risks and benefits of microbiota therapeutics. *Gastroenterology*. 2017;152(4):681–684.
16. Soldati L, Di Renzo L, Jirillo E, Ascierto PA, Marincola FM, De Lorenzo A. The influence of diet on anti-cancer immune responsiveness. *J Transl Med* [Internet]. 2018 Mar 20 [cited 2018 Aug 23];16. Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5859494/> PMID: PMC5859494
17. Balansky R, Ganchev G, Ilcheva M, Kratchanova M, Denev P, Kratchanov C, Polasa K, D'Agostini F, Steele VE, De Flora S. Inhibition of lung tumor development by berry extracts in mice exposed to cigarette smoke. *Int J Cancer*. 2012 Nov 1;131(9):1991–1997. PMID: 22328465
18. Aqil F, Jeyabalan J, Kausar H, Munagala R, Singh IP, Gupta R. Lung cancer inhibitory activity of dietary berries and berry polyphenolics. *Journal of Berry Research*. 2016 Jan 1;6(2):105–114.
19. Kausar H, Jeyabalan J, Aqil F, Chabba D, Sidana J, Singh IP, Gupta RC. Berry anthocyanidins synergistically suppress growth and invasive potential of human non-small-cell lung cancer cells. *Cancer Lett*. 2012 Dec 1;325(1):54–62. PMID: 22659736
20. Marín L, Miguélez EM, Villar CJ, Lombó F. Bioavailability of Dietary Polyphenols and Gut Microbiota Metabolism: Antimicrobial Properties. *Biomed Res Int* [Internet]. 2015 [cited 2018 Jun 6];2015. Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4352739/> PMID: PMC4352739
21. Anhê FF, Pilon G, Roy D, Desjardins Y, Levy E, Marette A. Triggering Akkermansia with dietary polyphenols: A new weapon to combat the metabolic syndrome? *Gut Microbes*. 2016 Feb 22;7(2):146–153. PMID: PMC4856456
22. Piening BD, Zhou W, Contrepois K, Röst H, Gu Urban GJ, Mishra T, Hanson BM, Bautista EJ, Leopold S, Yeh CY, Spakowicz D, Banerjee I, Chen C, Kukurba K, Perelman D, Craig C, Colbert E, Salins D, Rego S, Lee S, Zhang C, Wheeler J, Sailani MR, Liang L, Abbott C, Gerstein M, Mardinoglu A, Smith U, Rubin DL, Pitteri S, Sodergren E, McLaughlin TL, Weinstock GM, Snyder MP. Integrative Personal Omics Profiles during Periods of Weight Gain and Loss. *Cell Systems* [Internet]. 2018 Jan 17 [cited 2018 Feb 7]; Available from: <http://www.sciencedirect.com/science/article/pii/S2405471217305550>
23. Everard A, Belzer C, Geurts L, Ouwerkerk JP, Druart C, Bindels LB, Guiot Y, Derrien M, Muccioli GG, Delzenne NM. Cross-talk between Akkermansia muciniphila and intestinal epithelium controls diet-induced obesity. *Proceedings of the National Academy of Sciences*. 2013;110(22):9066–9071.
24. Casto BC, Kresty LA, Kraly CL, Pearl DK, Knobloch TJ, Schut HA, Stoner GD, Mallery SR, Weghorst CM. Chemoprevention of oral cancer by black raspberries. *Anticancer Res*. 2002 Dec;22(6C):4005–15.
25. Kresty LA, Frankel WL, Hammond CD, Baird ME, Mele JM, Stoner GD, Fromkes JJ. Transitioning from preclinical to clinical chemopreventive assessments of lyophilized black raspberries: interim results show berries modulate markers of oxidative stress in Barrett's esophagus patients. *Nutr Cancer*. 2006;54(1):148–56.
26. Mallery SR, Budendorf DE, Larsen MP, Pei P, Tong M, Holpuch AS, Larsen PE, Stoner GD, Fields HW,

Chan KK. Effects of human oral mucosal tissue, saliva and oral microflora on intraoral metabolism and bioactivation of black raspberry anthocyanins. *Cancer Prevention Research*. 2011;canprevres–0040.

27. Mallery SR, Stoner GD, Larsen PE, Fields HW, Rodrigo KA, Schwartz SJ, Tian Q, Dai J, Mumper RJ. Formulation and in-vitro and in-vivo evaluation of a mucoadhesive gel containing freeze dried black raspberries: implications for oral cancer chemoprevention. *Pharm Res*. 2007 Apr;24(4):728–37.
28. Stoner GD, Sardo C, Apseloff G, Mullet D, Wargo W, Pound V, Singh A, Sanders J, Aziz R, Casto B, Sun X. Pharmacokinetics of anthocyanins and ellagic acid in healthy volunteers fed freeze-dried black raspberries daily for 7 days. *J Clin Pharmacol*. 2005 Oct;45(10):1153–64.
29. Knobloch TJ, Uhrig LK, Pearl DK, Casto BC, Warner BM, Clinton SK, Sardo-Molmenti CL, Ferguson JM, Daly BT, Riedl K. Suppression of pro-inflammatory and pro-survival biomarkers in oral cancer patients consuming a black raspberry phytochemical-rich troche. *Cancer Prevention Research*. 2015;canprevres–0187.
30. Teegarden MD, Cichon MJ, Cooperstone JL, Ahn-Jarvis JH, Weghorst CM, Vodovotz Y, Schwartz SJ. Untargeted Profiling of the Urinary Metabolomes of Smokers and Nonsmokers after a Strawberry Intervention. *The FASEB Journal*. 2016;30(1_supplement):911–6.
31. Fisher EL, Ahn-Jarvis J, Gu J, Weghorst CM, Vodovotz Y. Assessment of physicochemical properties, dissolution kinetics and storage stability of a novel strawberry confection designed for delivery of chemopreventive agents. *Food Structure*. 2014;1(2):171–181.
32. Gu J, Ahn-Jarvis JH, Riedl KM, Schwartz SJ, Clinton SK, Vodovotz Y. Characterization of black raspberry functional food products for cancer prevention human clinical trials. *Journal of agricultural and food chemistry*. 2013;62(18):3997–4006.
33. Shiels MS, Katki HA, Freedman ND, Purdue MP, Wentzensen N, Trabert B, Kitahara CM, Furr M, Li Y, Kemp TJ. Cigarette smoking and variations in systemic immune and inflammation markers. *JNCI: Journal of the National Cancer Institute*. 2014;106(11).
34. Wellek S, Blettner M. On the Proper Use of the Crossover Design in Clinical Trials. *Dtsch Arztebl Int*. 2012 Apr;109(15):276–281. PMID: PMC3345345
35. A black raspberry extract inhibits proliferation and regulates apoptosis in cervical cancer cells. *Gynecologic Oncology*. 2011 Nov 1;123(2):401–406.
36. Kresty LA, Mallery SR, Stoner GD. Black raspberries in cancer clinical trials: Past, present and future. *J Berry Res*. 2016;6(2):251–261. PMID: PMC5008867
37. Bittoni MA, Focht BC, Clinton SK, Buckworth J, Harris RE. Prospective evaluation of C-reactive protein, smoking and lung cancer death in the Third National Health and Nutrition Examination Survey. *International journal of oncology*. 2015;47(4):1537–1544.
38. Screening FRSLC. U.S. Preventive Services Task Force [Internet]. December Page/Document/RecommendationStatementFinal/lung-cancer-screening; 2016. Available from: <https://www.uspreventiveservicestaskforce.org/>
39. Wang S, DeGross VL, Clinton SK. Tomato and soy polyphenols reduce insulin-like growth factor-I-stimulated rat prostate cancer cell proliferation and apoptotic resistance in vitro via inhibition of intracellular signaling pathways involving tyrosine kinase. *J Nutr*. 2003 Jul;133(7):2367–2376. PMID: 12840208
40. Kristal AR, Kolar AS, Fisher JL, Plascak JJ, Stumbo PJ, Weiss R, Paskett ED. Evaluation of Web-Based, Self-Administered, Graphical Food Frequency Questionnaire. *J Acad Nutr Diet*. 2014 Apr;114(4):613–621.

PMCID: PMC3966309

41. Bhagwat SH. USDA Database for the Flavonoid Content of Selected Foods. 2013.
42. Kato S, Berzofsky JA, Terabe M. Possible Therapeutic Application of Targeting Type II Natural Killer T Cell-Mediated Suppression of Tumor Immunity. *Front Immunol* [Internet]. 2018 Feb 22 [cited 2020 Apr 30];9. Available from: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5827362/> PMCID: PMC5827362
43. Szopinska JW, Gresse R, van der Marel S, Boekhorst J, Lukovac S, van Swam I, Franke B, Timmerman H, Belzer C, Arias Vasquez A. Reliability of a participant-friendly fecal collection method for microbiome analyses: a step towards large sample size investigation. *BMC Microbiology*. 2018 Sep 6;18(1):110.
44. Love MI, Huber W, Anders S. Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2. *Genome Biology*. 2014;15(12):550.
45. Kanehisa M, Araki M, Goto S, Hattori M, Hirakawa M, Itoh M, Katayama T, Kawashima S, Okuda S, Tokimatsu T, Yamanishi Y. KEGG for linking genomes to life and the environment. *Nucleic Acids Res*. 2008 Jan;36(Database issue):D480-484. PMID: 18077471
46. Luo W, Brouwer C. Pathview: an R/Bioconductor package for pathway-based data integration and visualization. *Bioinformatics*. 2013;29(14):1830–1831.
47. Hunter PR, Gaston MA. Numerical index of the discriminatory ability of typing systems: an application of Simpson's index of diversity. *Journal of Clinical Microbiology*. 1988 Nov 1;26(11):2465–2466. PMID: 3069867
48. Lozupone C, Lladser ME, Knights D, Stombaugh J, Knight R. UniFrac: an effective distance metric for microbial community comparison [Internet]. *The ISME Journal*. 2010 [cited 2018 Aug 23]. Available from: <https://www.nature.com/articles/ismej2010133>
49. Kruskal JB. Multidimensional scaling by optimizing goodness of fit to a nonmetric hypothesis. *Psychometrika*. 1964;29(1):1–27.
50. Mielke Jr PW, Berry KJ, Johnson ES. Multi-response permutation procedures for a priori classifications. *Communications in Statistics-Theory and Methods*. 1976;5(14):1409–1424.
51. Oksanen J, Blanchet FG, Friendly M, Kindt R, Legendre P, McGlinn D, Minchin PR, O'Hara RB, Simpson GL, Solymos P, Stevens MHH, Szoecs E, Wagner H. vegan: Community Ecology Package [Internet]. 2018. Available from: <https://CRAN.R-project.org/package=vegan>
52. Spakowicz D, Lou S, Barron B, Li T, Chupp G, Gerstein M. Approaches for integrating heterogeneous RNA-seq data reveals cross-talk between microbes and genes in asthmatic patients. *bioRxiv*. 2019;
53. Chang J. lda: Collapsed Gibbs Sampling Methods for Topic Models [Internet]. 2015. Available from: <https://CRAN.R-project.org/package=lda>
54. Alexa A, Rahnenfuhrer J. topGO: Enrichment Analysis for Gene Ontology. 2018.
55. Prestat E, David MM, Hultman J, Taş N, Lamendella R, Dvornik J, Mackelprang R, Myrold DD, Jumpponen A, Tringe SG, Holman E, Mavromatis K, Jansson JK. FOAM (Functional Ontology Assignments for Metagenomes): a Hidden Markov Model (HMM) database with environmental focus. *Nucleic Acids Res*. 2014 Oct 29;42(19):e145–e145.

13. Appendices